Issue 1: Define

(a) Alphabet: as finite, non-Empty Set of elements denoted by Σ which contains any symbols or characters

Eg: +, *, ., -, a to z.

(b) Power of an alphabet: Is the set of all possible strings of specified length formed using the symbol denoted as Σ^n

Eg: $\Sigma^0 = \epsilon$, $\Sigma^1 = 1$

(c) Symbols: A symbol is a single, indivisible character or basic unit used to form strings

Eg: 1, 0, a, b...

(d) Strings: The finite sequence of symbols taken from an alphabet

Eg: aa, abc...

(e) Languages: Set of strings formed using an alphabet formally a language L is any subset of Σ^*

Eg: $L = \{a, aaa, aaaaa...\}$

(f) DFA: Deterministic finite State Machine is a computational Device whose input is string & output is accept or reject

$M = (Q, \Sigma, \delta, q_0, F)$: Quin tuples

$Q \rightarrow$ finite set of states $q_0 \rightarrow$ Start state $q_0 \in Q$

$\Sigma \rightarrow$ Input alphabets $F \rightarrow$ set of final state $F \subseteq Q$

$\delta \rightarrow$ Transition Function which maps from $Q \times \Sigma$ to Q

(g) NFA: Non Deterministic finite Automata is a computational device which input is a string & output is accept or reject. For a given input it may have zero, one or multiple possible transitions from a state.

$M = (Q, \Sigma, \delta, q_0, F)$ → Quin tuples

Q → finite set of states q_0 → initial state $q_0 \in Q$

Σ → input alphabets F → Set of final state $F \subseteq Q$

δ → Transition function $\delta : Q \times \Sigma \rightarrow 2^Q$

(2) Define Automata Theory? Explain why do we need it.

Automata Theory: branch of Theoritical Computer Science that deals with the study of abstract computational machines called automata & their ability to perform computations & recognise languages.

Why do we need?

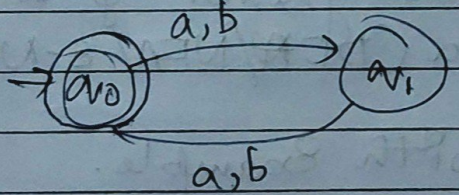
- To understand computation
- To design FSM
- To recognize patterns
- To design compilers
- To study computational limitations
- To model real-world systems

} elaborate in one line each depending on marks

(3) With an example explain when we say that a language is accepted by a machine?

L is accepted by a machine M if it reaches the final state after processing the entire input string

$L(M) = \{w / w \in \{a,b\}$ where w is even length $\}$



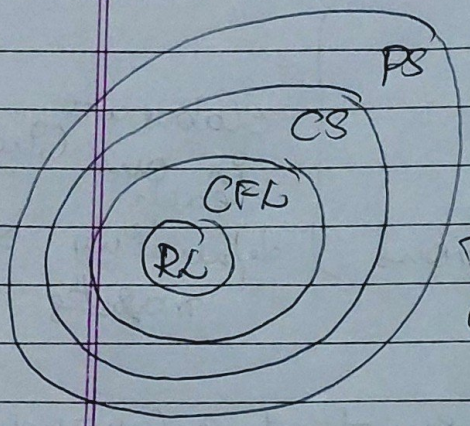
$L = \{\varepsilon, ab, aa, bb, ba \ldots\}$

(4) Explain Applications of Automata theory?

Ans (a) Compiler Design : lexical analysis
(b) Text processing : for pattern matching & searching
(c) Digital circuit Design : Sequential circuits & control
(d) Natural Language processing
(e) Protocol design
(f) Vending Machines : traffic lights, other control systems
(g) Computability Theory : computational machines

(5) With neat diagram Explain Hierarchy of Language Class

Ans



Type 0 : phase structure Language
→ Language generated by unrestricted grammer & accepted by Turing Machine (TM)

Type 1 : Context-Sensitive
→ L generated by context-sensitive Grammer & accepted by Linear Bounded Automata

Type 2 : Context free Language
→ L generated by context free Grammer & accepted by pushdown Automata

Type 3 : Regular Language
→ L generated by Regular grammer & accepted by a Finite Automata (DFA/NFA/ε-NFA)

(6) DFA, NFA, ε-NFA with Example.

Ans Define DFA NFA & ε-NFA
ε-NFA : Allows transitions from one state to another that any input symbol using ε-transitions
$M = (Q, \Sigma, \delta, q_0, F)$

$$\delta : Q \times (\Sigma \cup \{\varepsilon\}) \longrightarrow 2^Q$$

(7) Contrast between DFA NFA ε-NFA

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	DFA	NFA	ε-NFA
(a)	5-tuple $M=(Q,\Sigma,\delta,q_0,F)$ $Q \rightarrow$ set of states $\Sigma \rightarrow$ input alphabets $\delta: Q \times \Sigma \rightarrow Q$ q_0: initial state $F \subseteq Q$: final states	$\delta: Q \times \Sigma \rightarrow 2^Q$	$\delta: Q \times (\Sigma \cup \{\varepsilon\}) \rightarrow 2^Q$
(b)	There is zero or one transition from a state	There is zero, one or more	There can be zero, one or more also with ε
(c)	more no of transitions	less no.	Relatively more than NFA
(d)	difficult to construct	Easy to construct	Easy using Regular Expressions
(e)	Less powerful	more powerful	more powerful than NFA

TOC module - 1 QB

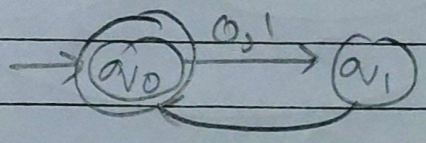
DFA → for all answers write $Q, \Sigma, \delta, F, q_0$

δ → table

(8) $\Sigma = \{0, 1\}$ draw DFA

(i) $L = \{w \mid w \text{ is even length}\}$ for even number

ε, 00, 01, 10, 11, ...



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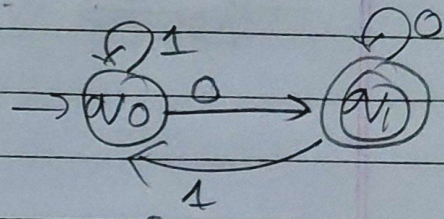
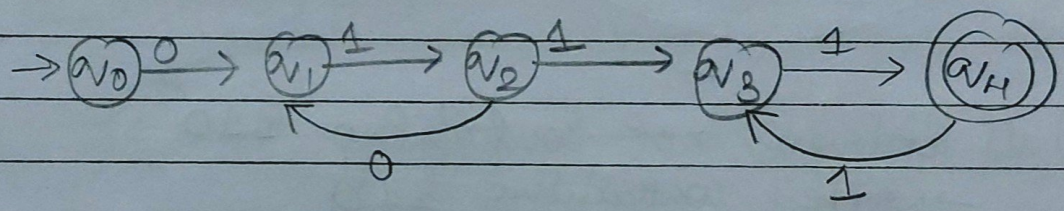


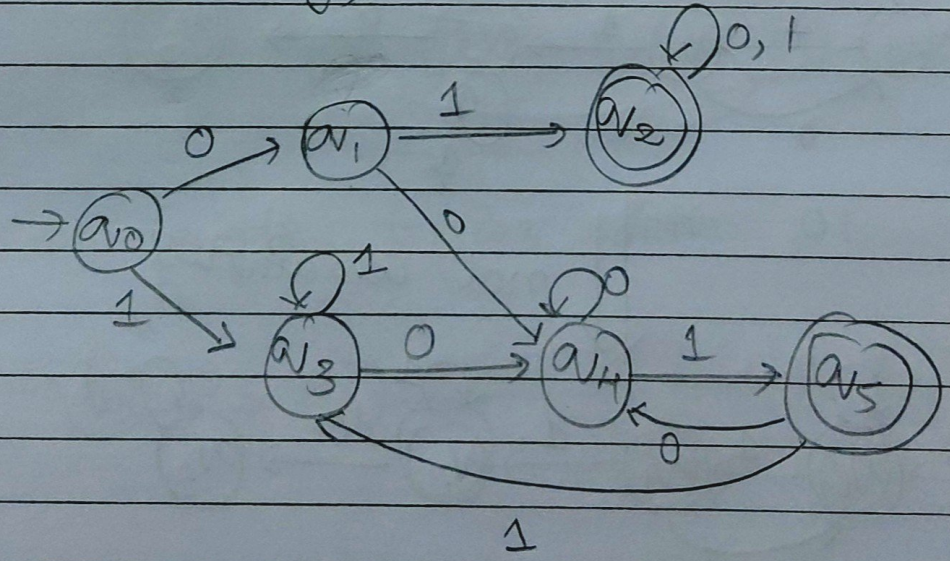
(ii) $L = \{(01)^i (1)^{2j} \mid i >= 1, j >= 1\}$

$L = \{0111,$

0101011111



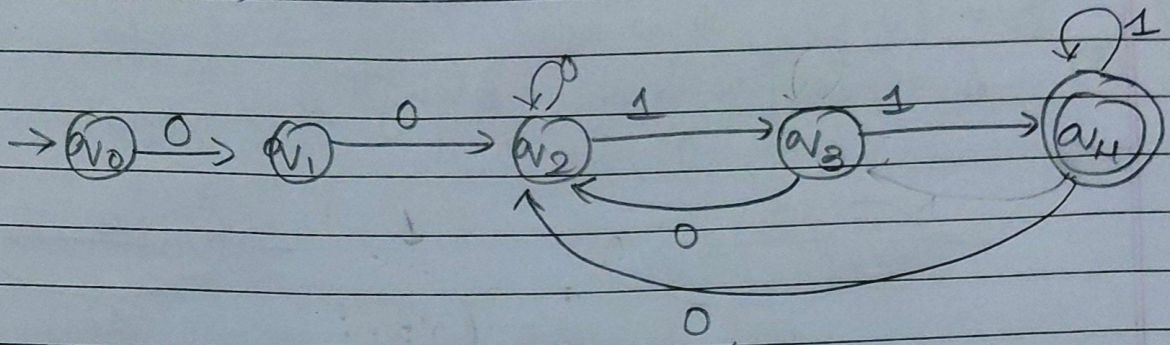
(iii) 01 → beg, end or both



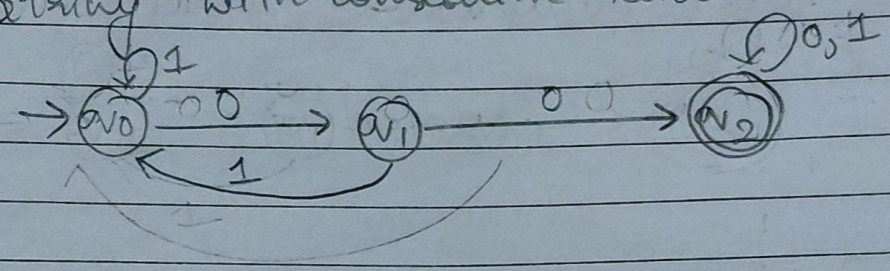
(iv) at least 2 zeros starting & ending with 2 1's

$0^i 1^j \quad i \geq 2, j \geq 2$

0011, 00000011

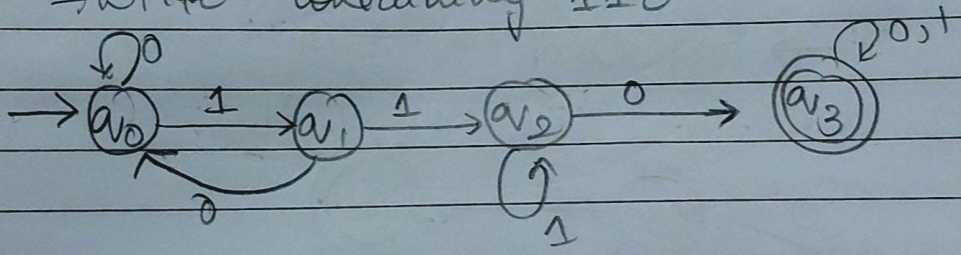


(v) String with consecutive zeros

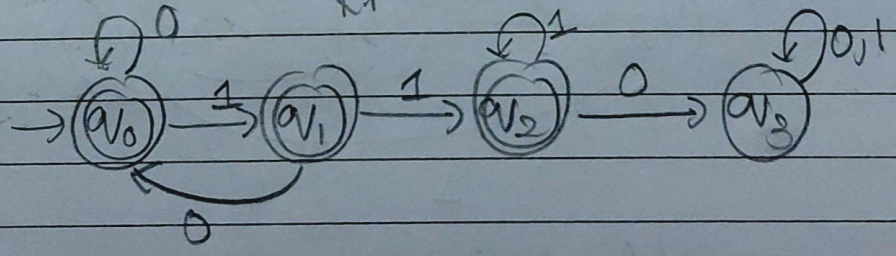


(vi) all strings not containing 110 aab

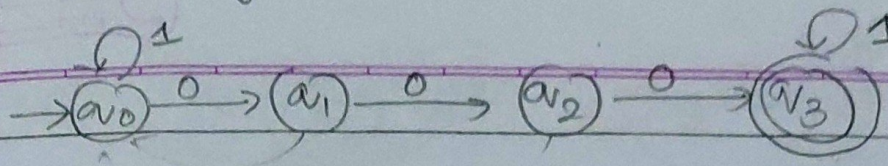
→ write containing 110



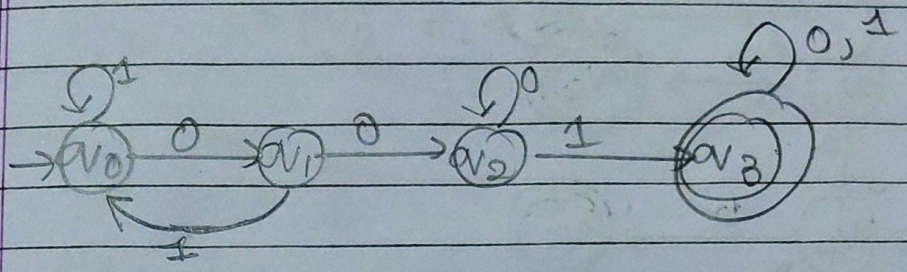
not containing



(vii) Set of strings exact 3 zeros



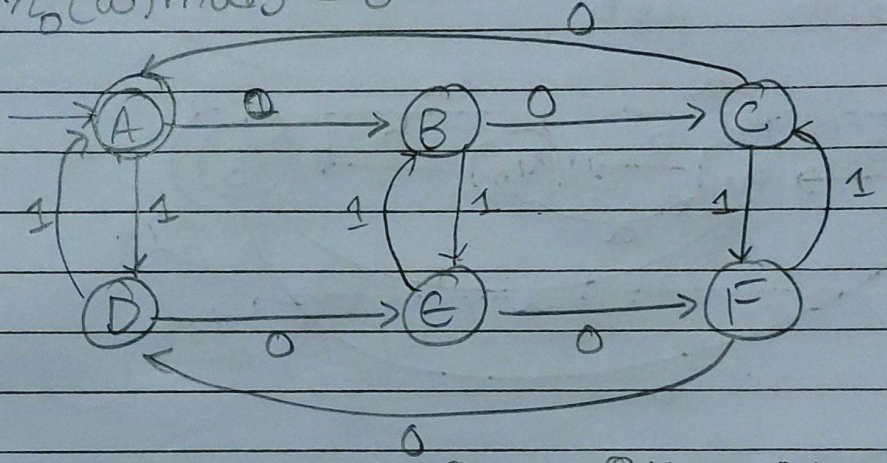
(viii) Substring 001



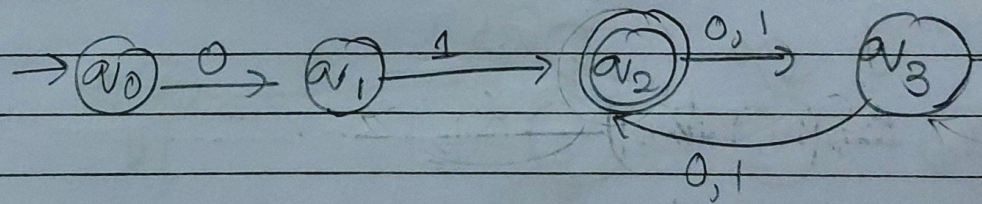
(ix) 1's are even 0's are multiple of 3

$n_1(w) \bmod 2 = 0$

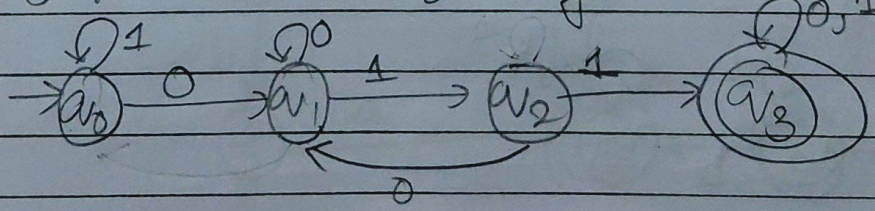
$n_0(w) \bmod 3 = 0$



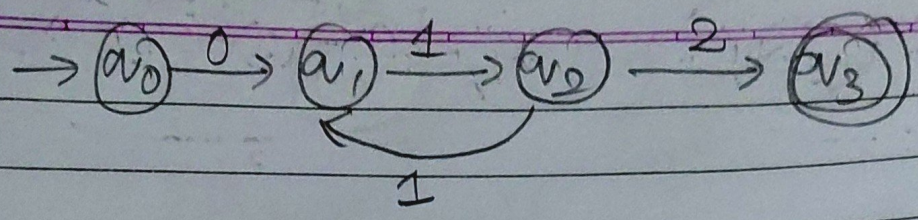
(x) w even length begins with 01



(xi) 0 11 as Substring

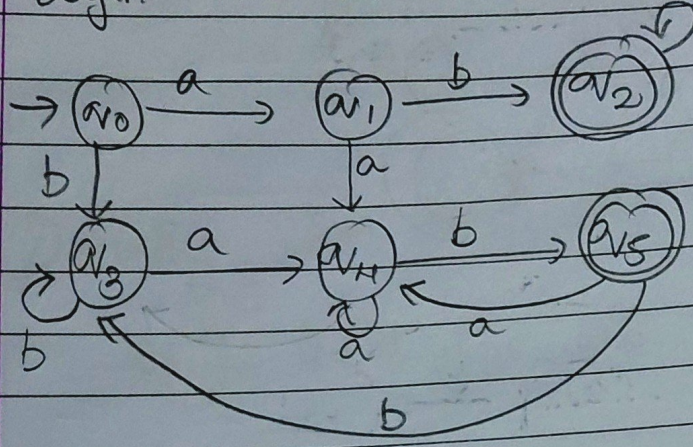


(xii) $0(1)^n 2$ n is odd

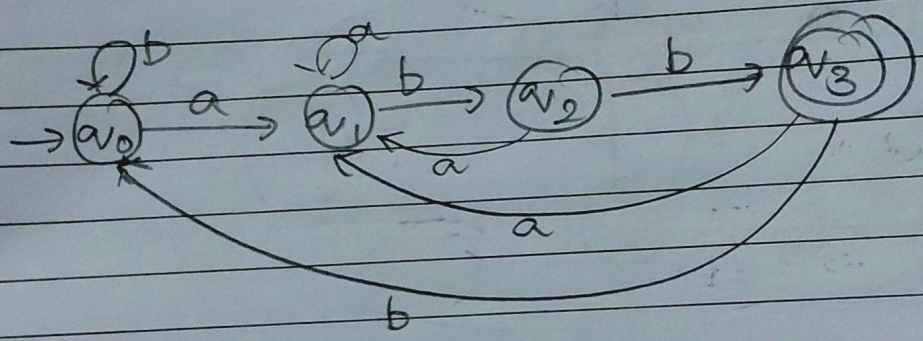


(g) $\Sigma = \{a, b\}$

(i) begin end or both with ab

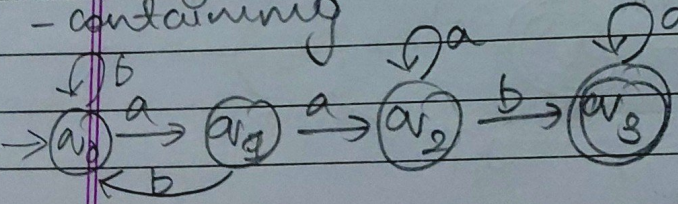


(ii) ends with abb

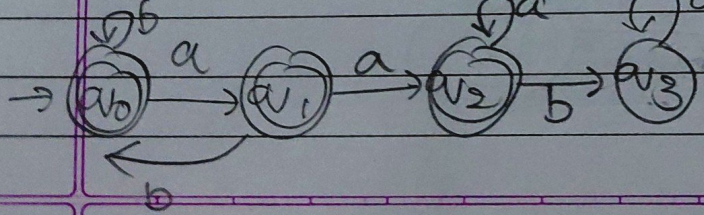


(iii) not containing aab

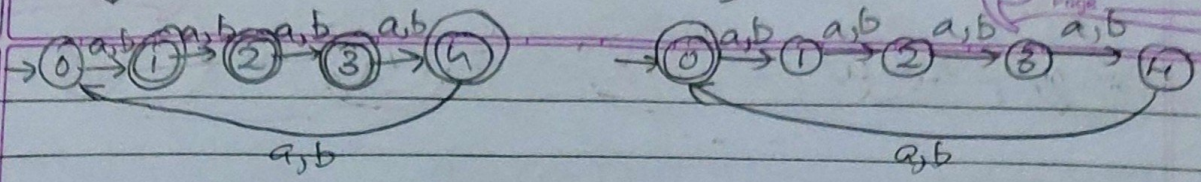
- containing



- not containing

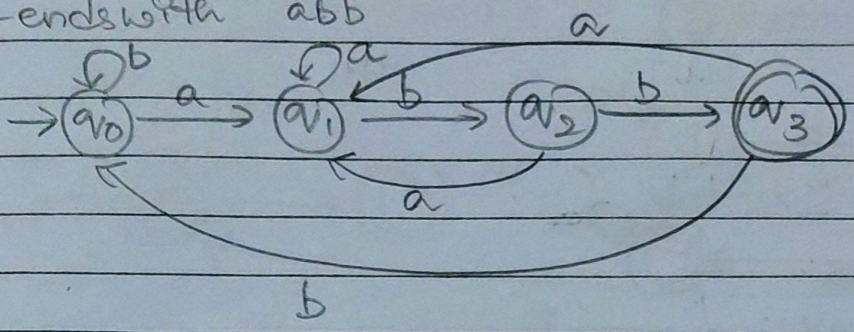


(iv) $|w| \bmod 5 \neq 0$ or $|w| \bmod 5 = 0$

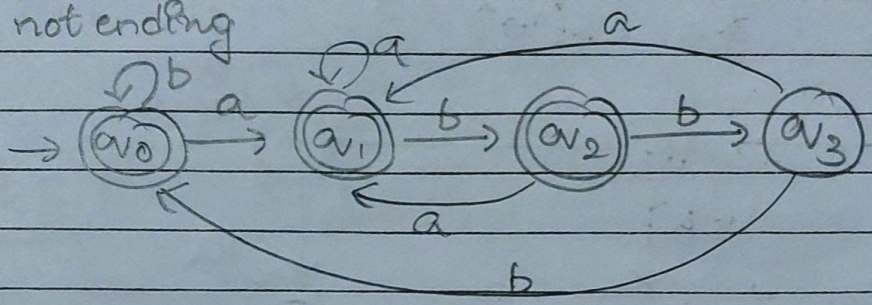


(v) do not ends with abb

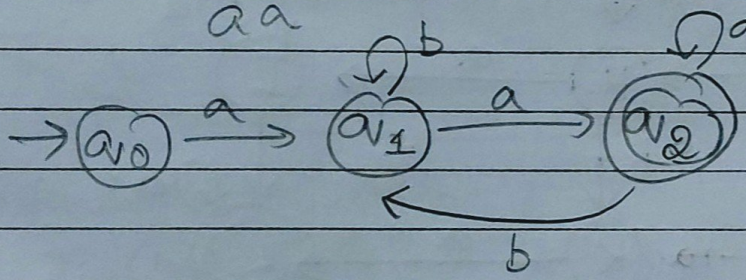
- ends with abb



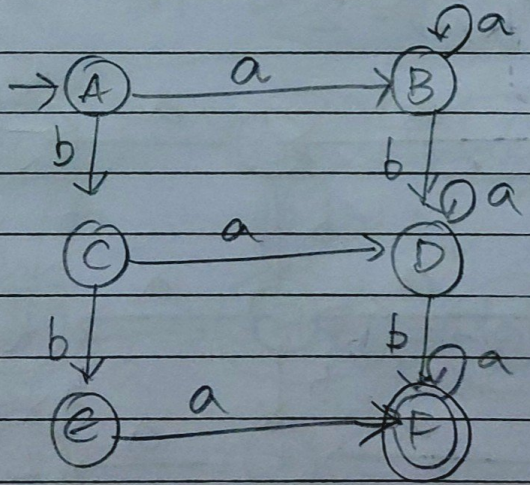
- not ending



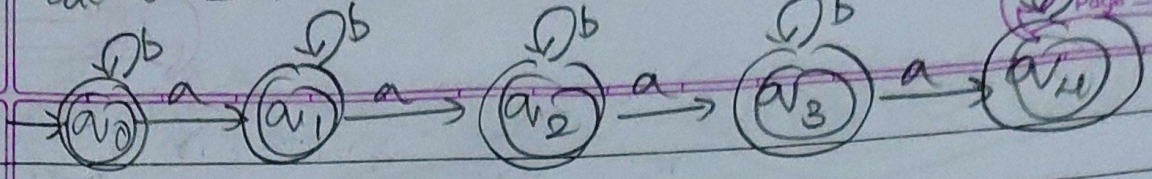
(vi) $awa \mid w \in \{a,b\}^*$



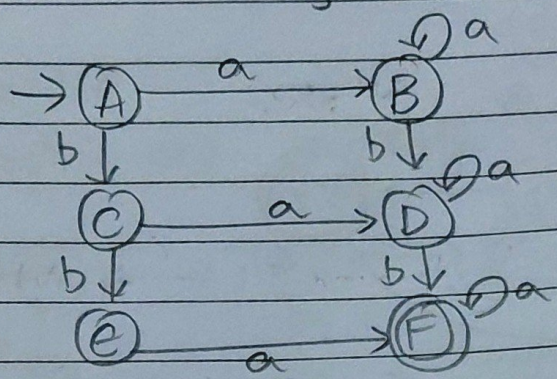
(vii) $n_a(w) \geq 1$ $n_b(w) = 2$



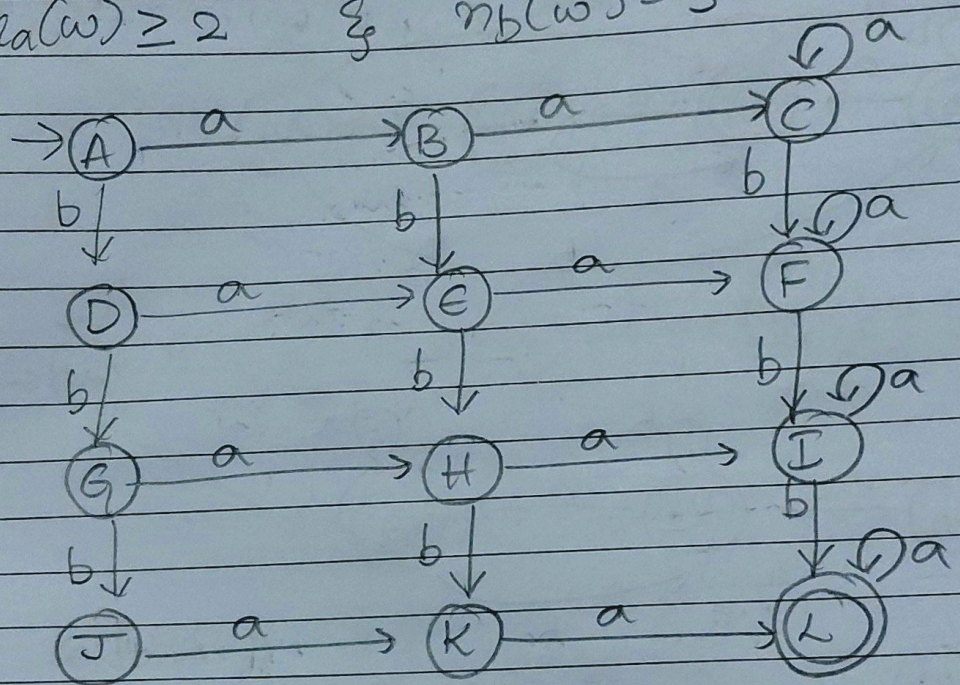
(viii) $n_a(w) \leq 4$



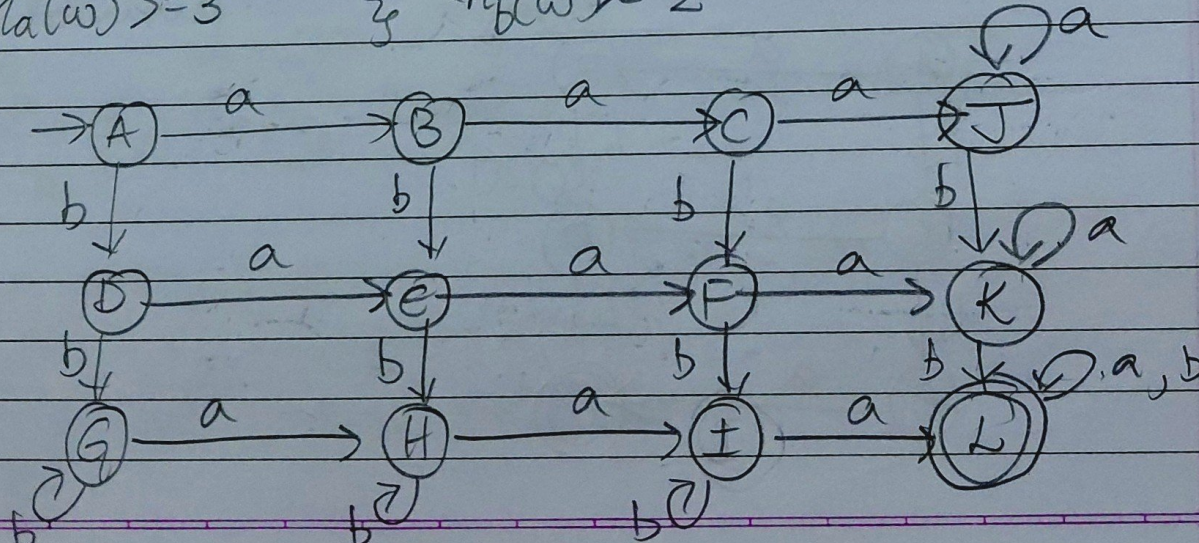
(ix) $n_a(w) \geq 1$ & $n_b(w) = 2$



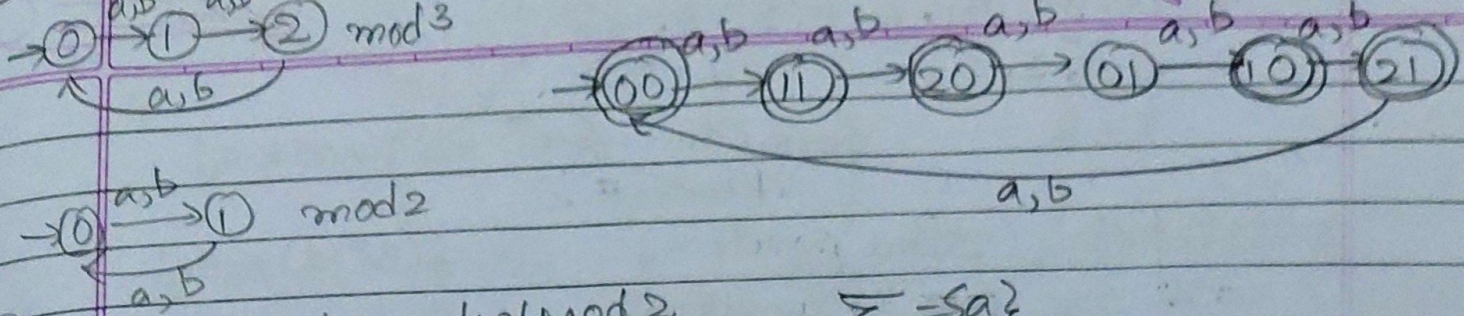
(x) $n_a(w) \geq 2$ & $n_b(w) = 3$



(xi) $n_a(w) >= 3$ & $n_b(w) >= 2$

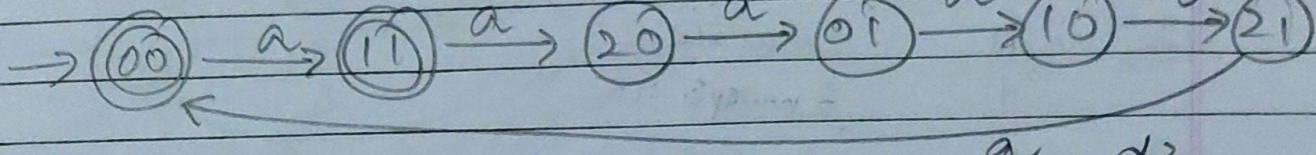


(xii) $|w| \bmod 3 >= |w| \bmod 2$



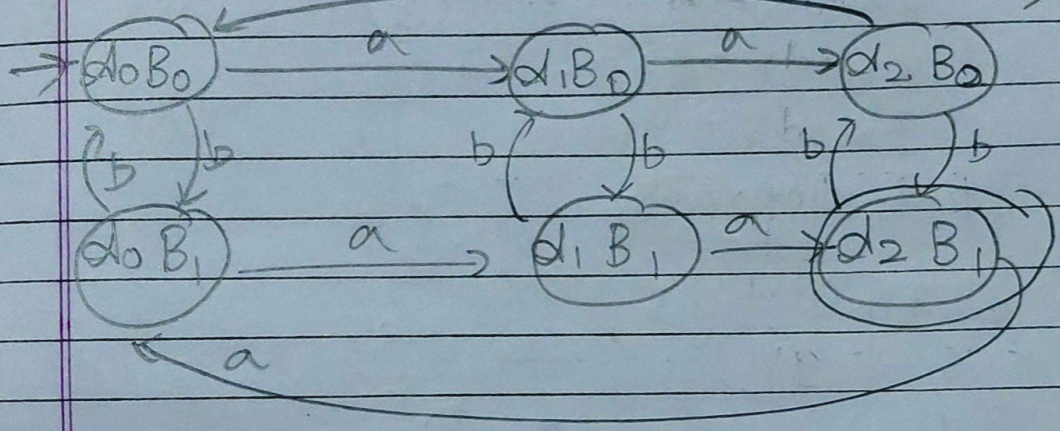
(xiii) $|w| \bmod 3 = |w| \bmod 2$ $\quad \Sigma = \{a\}$

00 & 11



(xiv) $L = \{ w \mid w \in (a+b)^* \quad n_a(w) \bmod 3 = 2 \leftarrow d_2 \quad n_b(w) \bmod 2 = 1 \leftarrow B_1$

$d_0, d_1, d_2 \leftarrow$ $\quad \hookrightarrow B_0, B_1$

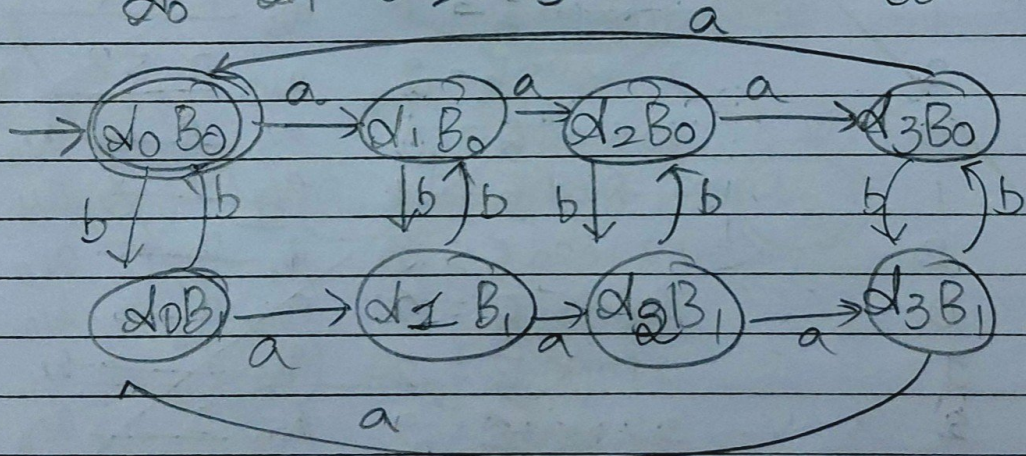


d_2B_1 ↓ final state

(xv) Extended transition ?

(xvi) $n_a(w)$ divisible by 4 $\quad n_b(w)$ divisible by 2

d_0 d_1 d_2 d_3 $\quad B_0$ B_1



(10) DFA to accept binary {0, 1}

(a) divisible by 5

$k = 5$ $i = \{0, 1, 2, 3, 4\}$

$d = \{0, 1\}$

$r = 2$ formula $(r * i + d) \bmod k = j$

$\delta(a_i, d) = a_j$

i	input d	$(r*i+d) \bmod k = j$
0	0	$j = 0$
	1	$j = 1$
1	0	$2 \bmod 5 = 2$
	1	$3 \bmod 5 = 3$
2	0	$4 \bmod 5 = 4$
	1	$5 \bmod 5 = 0$
3	0	$6 \bmod 5 = 1$
	1	$7 \bmod 5 = 2$
4	0	$8 \bmod 5 = 3$
	1	$9 \bmod 5 = 4$

$a_0 \xrightarrow{0} a_0$

$a_0 \xrightarrow{1} a_1$

a_1

a_1

So on

put diagram

(11) decimal divisible by 5

0 → 3 → 6 9

1 4 → 7

2 5 8

$r = 10$

check for only 3

0	0	$0 \bmod 5 = 0$
	1	$1 \bmod 5 = 1$
	2	$2 \bmod 5 = 2$
1	0	$10 \bmod 5 = 0$
	1	$11 \bmod 5 = 1$
	2	$12 \bmod 5 = 2$
2	0	$20 \bmod 5 = 0$
	1	$21 \bmod 5 = 1$
	2	$22 \bmod 5 = 2$

$a_0 \xrightarrow{0\,3\,6\,9} a_0$

$a_0 \xrightarrow{1\,4\,7} a_1$

$a_0 \xrightarrow{2\,5\,8} a_2$

$a_1 \xrightarrow{0} a_0$

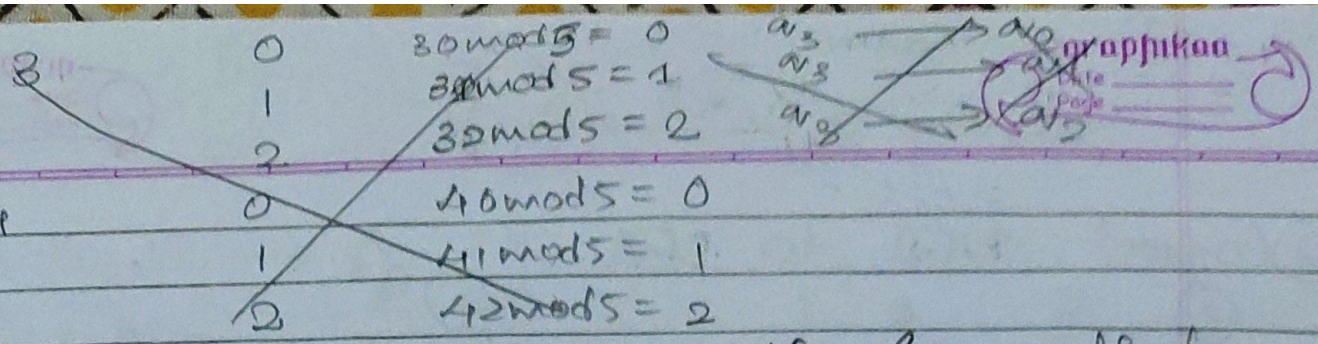
$a_1 \xrightarrow{1} a_1$

$a_1 \xrightarrow{2} a_2$

$a_2 \xrightarrow{0} a_0$

$a_2 \xrightarrow{1} a_1$

$a_2 \xrightarrow{2} a_2$

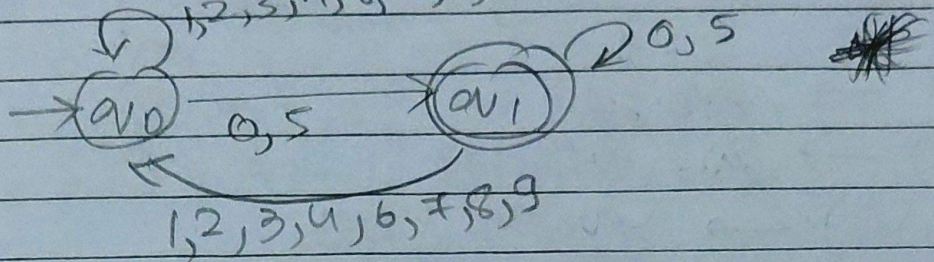


0 or 5 → a_0

1 2 3 4 6 7 8 9 → a_1

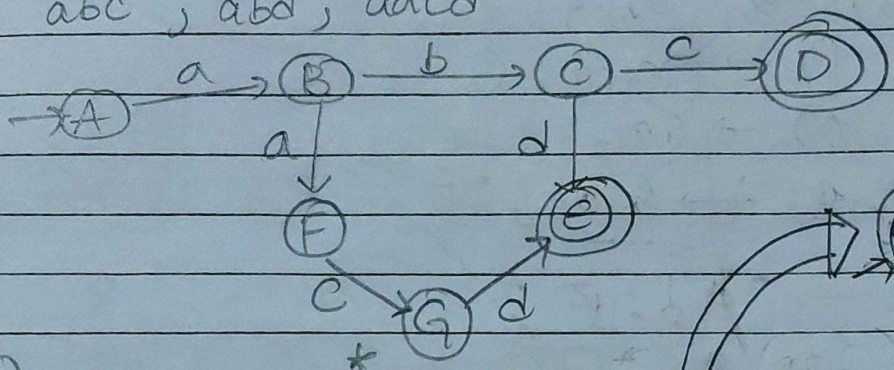
do the same method we get the answer

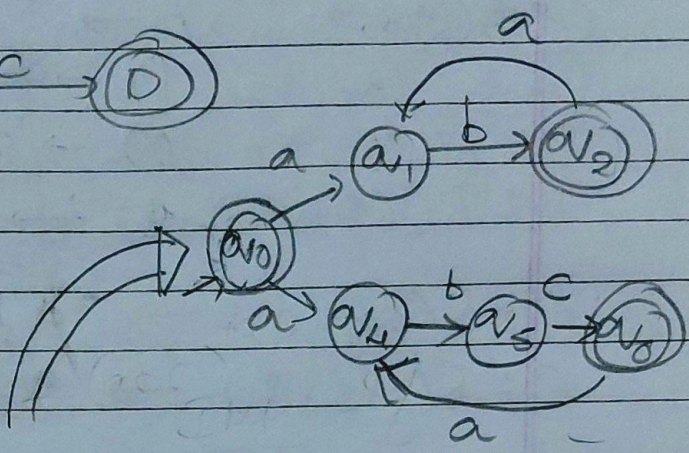
1,2,3,4,6,7,8,9



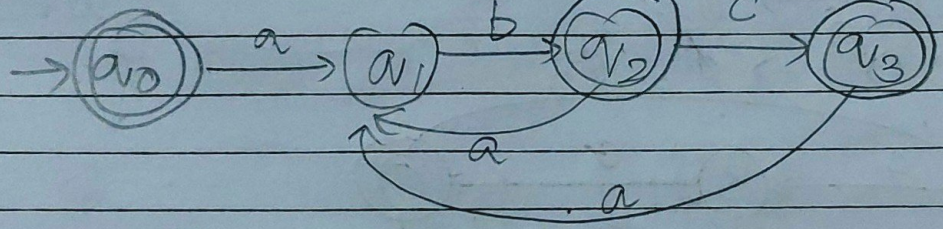
(II) ε-NFA or NFA

(a) abc , abd , aacd

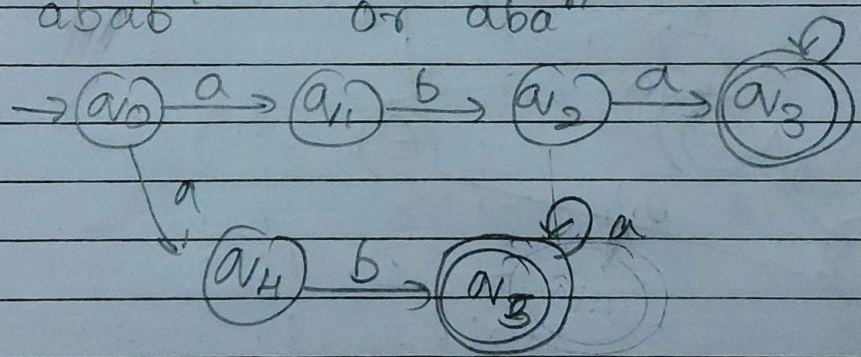




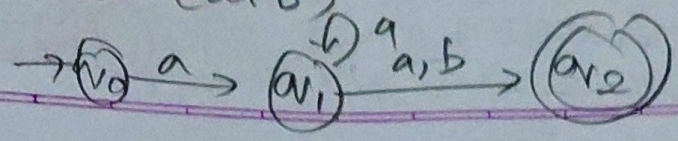
(b) $\{ab, abc\}^*$



(c) $ab(ab)^n$ or $ab(a)^n$



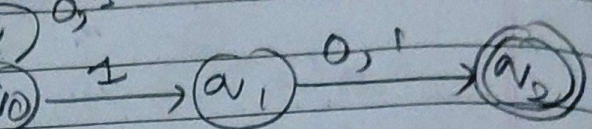
(1) $aa^*(a+b)$



(2) Convert NFA to DFA

(a)

δ	0	1
$\rightarrow q_0$	q_0	q_0, q_1
q_1	q_2	q_2
$*q_2$	ϕ	ϕ



Lazy Evalve method

A: $\delta(q_0, 0) \rightarrow q_0 : A$

A: $\delta(q_0, 1) \rightarrow \{q_0, q_1\} : B$

B: $\delta((q_0, q_1), 0) \rightarrow \{q_0, q_2\} : C$

B: $\delta((q_0, q_1), 1) \rightarrow \{q_1, q_2\} : D$

C: $\delta((q_0, q_2), 0) \rightarrow \{q_0\} : A$

C: $\delta((q_0, q_2), 1) \rightarrow \{q_1\} : E$

D: $\delta((q_1, q_2), 0) \rightarrow \{q_2\} : F$

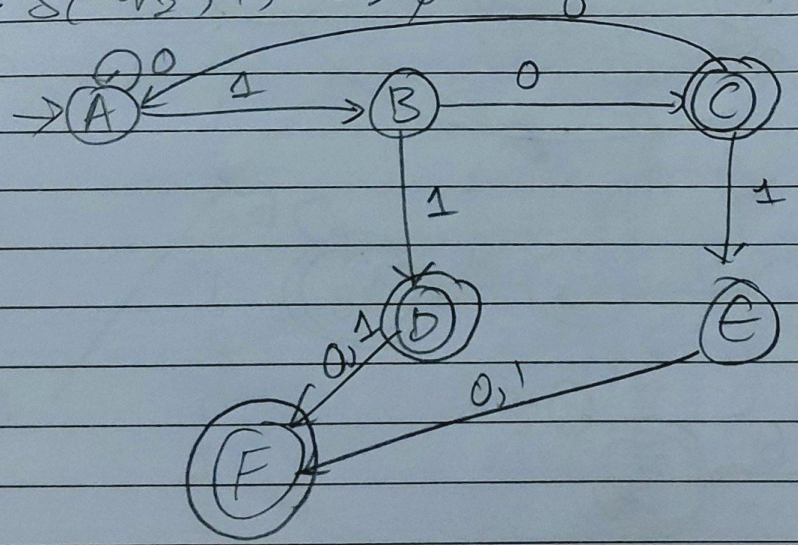
D: $\delta((q_1, q_2), 1) \rightarrow \{q_2\} : F$

E: $\delta(q_1, 0) \rightarrow q_2 : F$

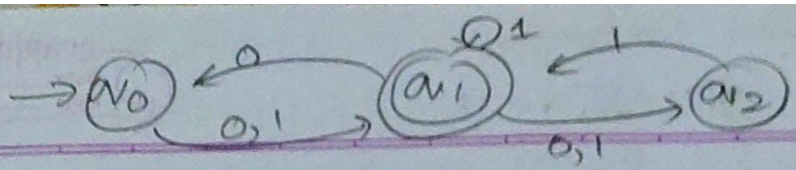
E: $\delta(q_2, 1) \rightarrow q_2 : F$

F: $\delta(q_2, 0) \rightarrow \phi$

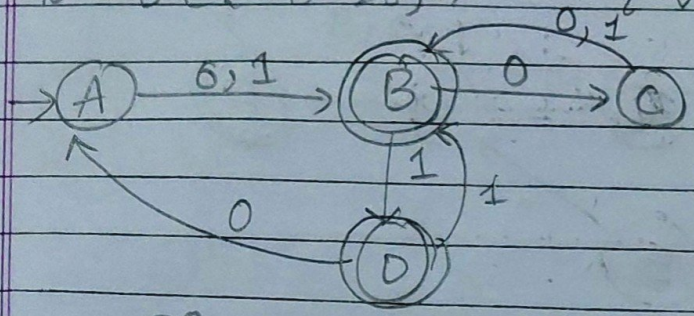
F: $\delta(q_2, 1) \rightarrow \phi$



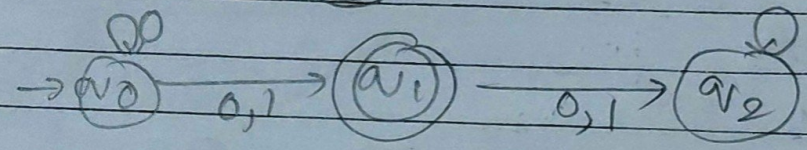
(b)



$A: \delta(q_0, 0) \rightarrow q_1 : B$
$A: \delta(q_0, 1) \rightarrow q_1 : B$
$B: \delta(q_1, 0) \rightarrow (q_0, q_2) : C$
$B: \delta(q_1, 1) \rightarrow (q_1, q_2) : D$
$C: \delta((q_0, q_2), 0) \rightarrow \{q_1\} : B$
$C: \delta((q_0, q_2), 1) \rightarrow \{q_1\} : B$
$D: \delta(\{q_1, q_2\}, 0) \rightarrow \{q_0\} : A$
$D: \delta(\{q_1, q_2\}, 1) \rightarrow \{q_1\} : B$



(c)



Simple

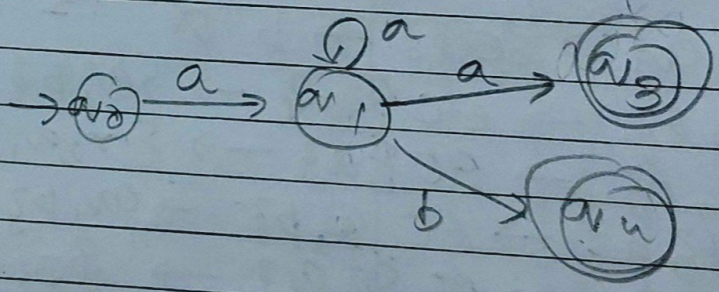
(d)

δ	a	b
$\rightarrow q_0$	q_0, q_1	q_0
q_1	–	q_3
$*q_2$	–	–

Q^n Incorrect

(e) & (f) → Same as previous use Lazy Evaluation

$aa^*a + ac^*b$



(13) (a) Subset construction scheme

P Q R S

Subset ϕ, P, Q, R, S

(P,Q), (P,R), (P,S)

(Q,R), (Q,S)

(R,S)

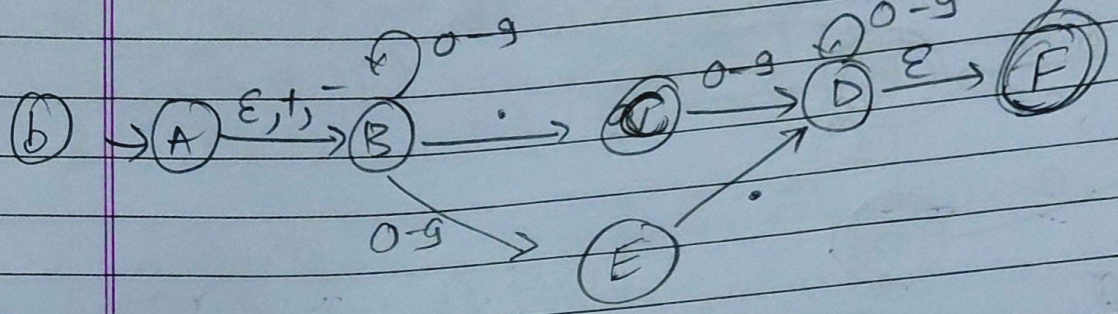
(P,Q,R), (P,Q,S)

(P,R,S), (Q,R,S)

(P,Q,R,S)

→ for all these we have to find the ε-cln

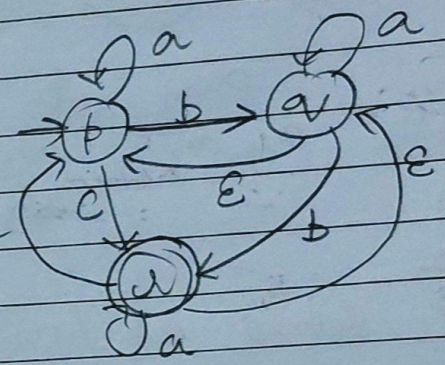
(b)



(14) ε-NFA to DFA

via ε-closure – εC

		ε	a	b	c
(a)		p	q	r	s
(b)	→ p	ϕ	p	q	r
	q	b	q	r	ϕ
Same	*r	q	r	ϕ	p



$\varepsilon C(p) = \{p\}$

$\varepsilon C(q) = \{q, p\}$

$\varepsilon C(r) = \{r, q, p\}$

A: $\varepsilon C\{\{p\}, a\} \rightarrow \varepsilon C\{p\} \rightarrow p$ A

A: $\varepsilon C\{\{p\}, b\} \rightarrow \varepsilon C\{q\} \rightarrow \{q, p\}$ B

A: $\varepsilon C\{\{p\}, c\} \rightarrow \varepsilon C\{r\} \rightarrow \{r, q, p\}$ C

B: $\varepsilon C\{\{q, p\}, a\} \rightarrow \varepsilon C\{q, p\} \rightarrow \{q, p\}$ B

B: $\varepsilon C\{B, b\} \rightarrow \varepsilon C\{r, q\} \rightarrow \{p, q, r\}$ C

B: $\varepsilon C\{B, c\} \rightarrow \varepsilon C\{r\} \rightarrow \{p, q, r\}$ C

C : ∈C { } {p, q, r}, a} → C{p, q, r} → C

C : ∈C { } {p, q, r}, b} → ∈C{q, r} → {p, q, r} ← C

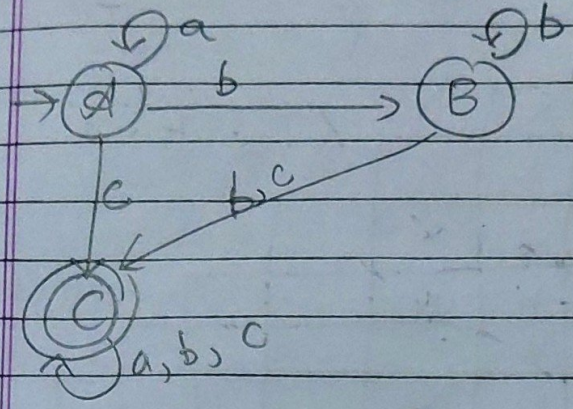
C : ∈C { } {p, q, r}, c} → ∈C{q, p} → {p, q, r} → C

D : ∈C { } {q, r}, a} → ∈C{q, r} → D

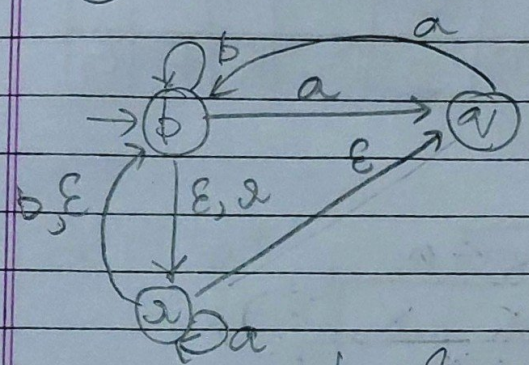
D : ∈C { } {q, r}, b} → ∈C{r} → {p, q, r} C

D : ∈C { } {q, r}, c} → ∈C{p} → p ← A

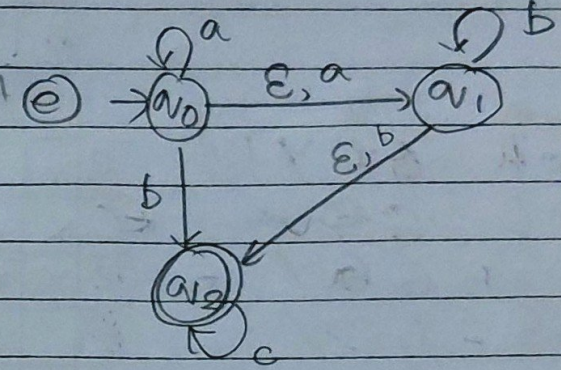
E : ∈C { } {q, p} a} → ∈C{q, p} →



(d)



(e)



(b) Same way

(15) Minimize dfa

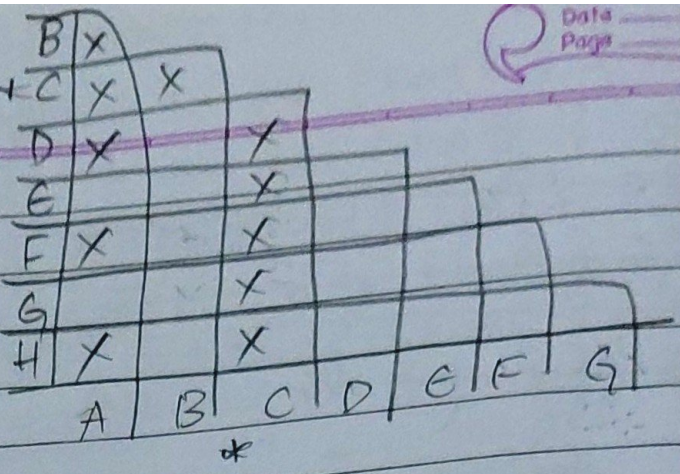
(a)

	ε	a	b
→A	B	F
B	G	C
*C	A	C
D	C	G
E	H	F
F	C	G
G	G	E
H	G	C

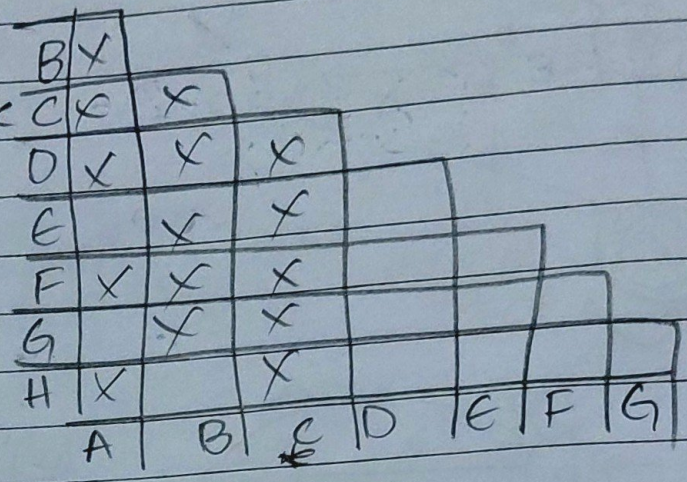
	A	B	C	D	E	F	G
B							
*C	X	X					
D			X				
E			X				
F			X				
G			X				
H			X				

*

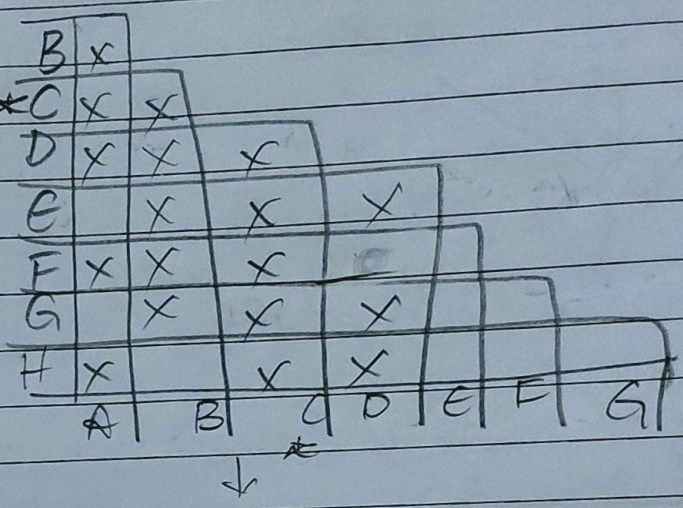
	a	b
A B	B G	F C X
~~A D~~	B C	F G X
A E	B H	F F
A F	B C	F G X
A G	B G	F E
A H	B G	F C X



	a	b
B, D	G C	C G X
B, E	G H	C F X
B, F	G C	C G X
B, G	G G	C E X
B, H	G G	C C ✓



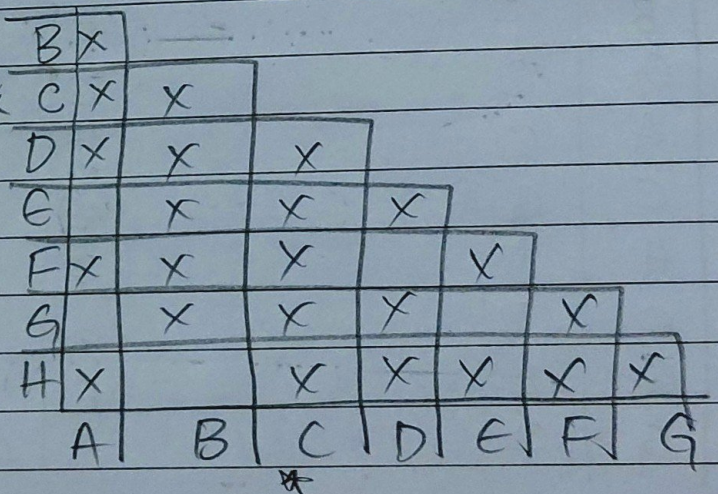
	a	b
D E	C H	G F X
D F	C C	G G ✓
D G	C G	G E X
D H	C G	G C X



	a	b
E F	H C	F G X
E, G	H G	F G ✓
E, H	H G	F C X

	a	b
F, G	C G	G E X
F, H	C G	G C X

	a	b
G H	G G	E C X



A,E	BH	FF ✓
A,G	BG	FE ✗
B,H	GG	CC ✓
D,F	CC	GG ✓
E,G	HG	FE ✗

(A,E) , (B,H) , (D,F) , C , G

TOC Module-2 QB

(1) → Application of RE
- → Lexical analysis
- → Pattern Matching
- → Text Editing
- → Input validation
- → Text processing
- → Data Extraction
- → Syntax Highlighting
- → Log file analysis

Operations

., *, +, ?, ^, $, [ ], [^], \, (), \|

- ? → 0/1
- ^ → beg of the
- $ → end of line
- [ ] → any one char within brackets
- [^] → except those inside braces
- \ → remove special many char
- () → group together
- \| → OR between Exp

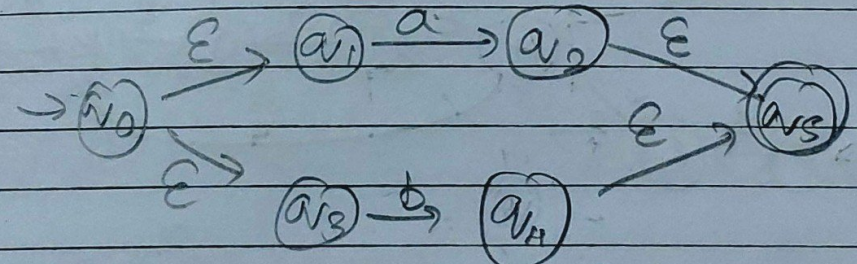
Ex: a*, a|+, a|?, ., ^a, b$, a\|b

[0-9], [a-z], [^a], a\{3\}, (a+b)

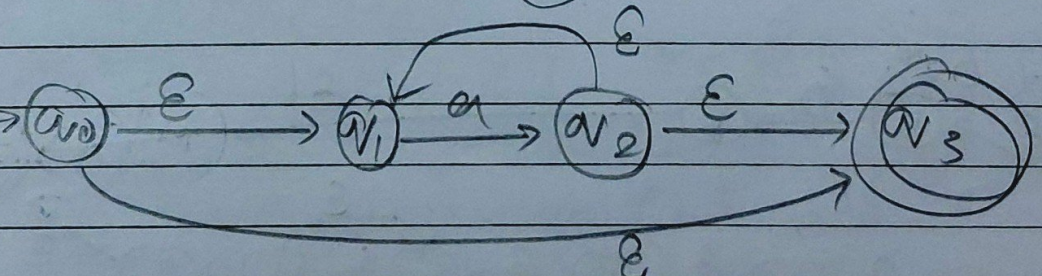
(2) To obtain ε-NFA 3 rulz are needed.

a.b → $q_0 \xrightarrow{a} q_2 \xrightarrow{\varepsilon} q_3 \xrightarrow{b} q_4$

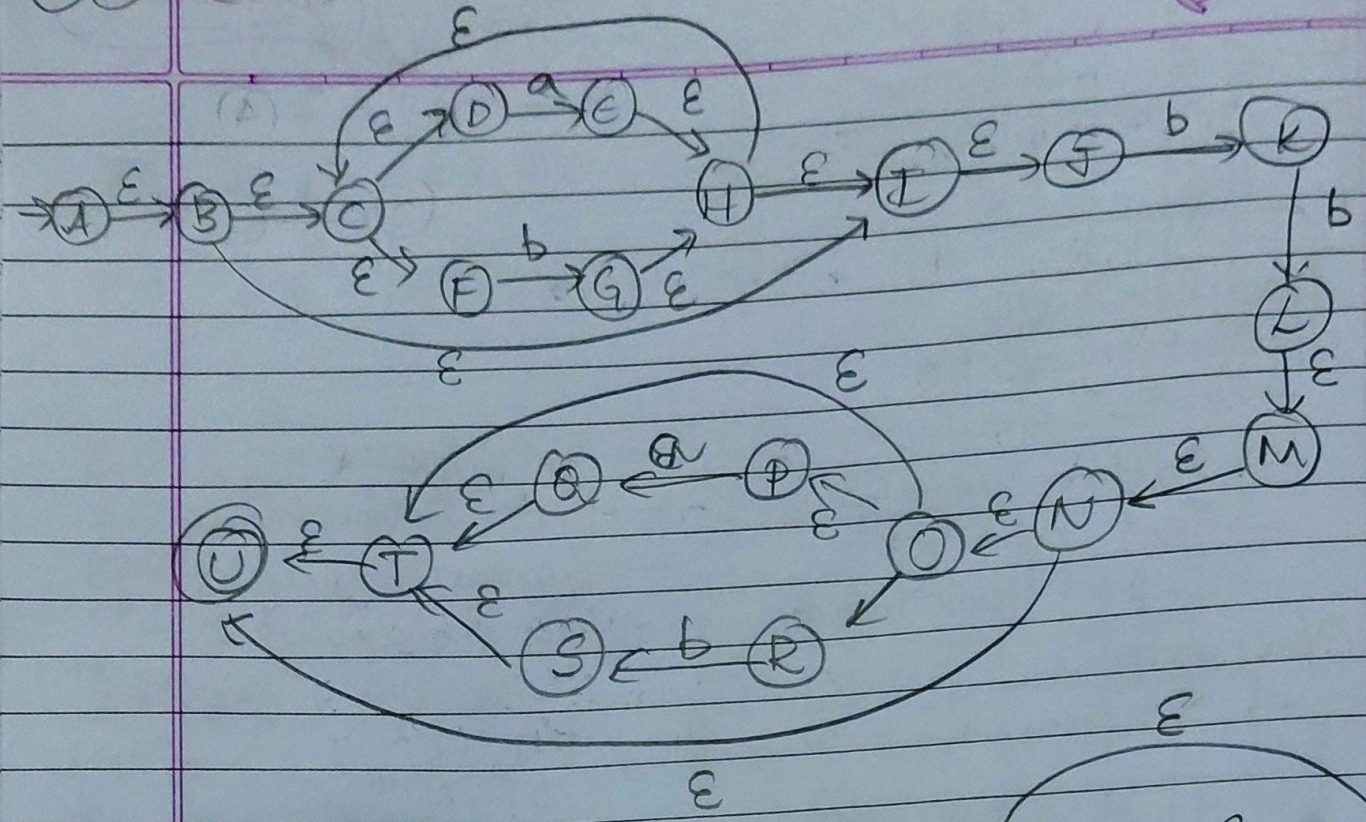
a+b



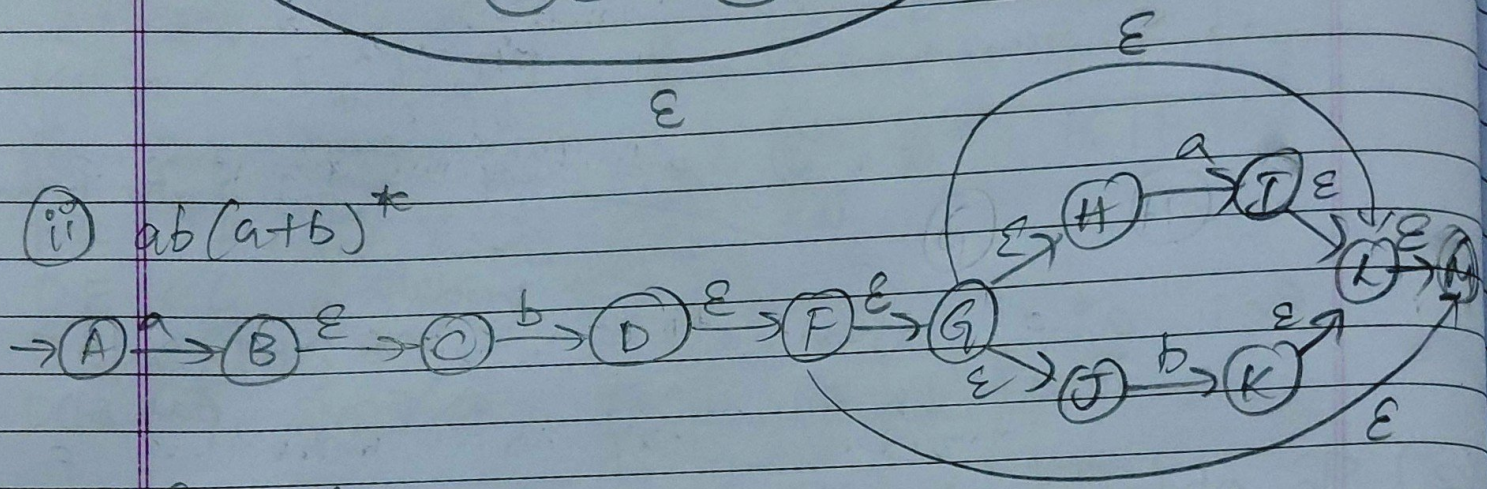
a*



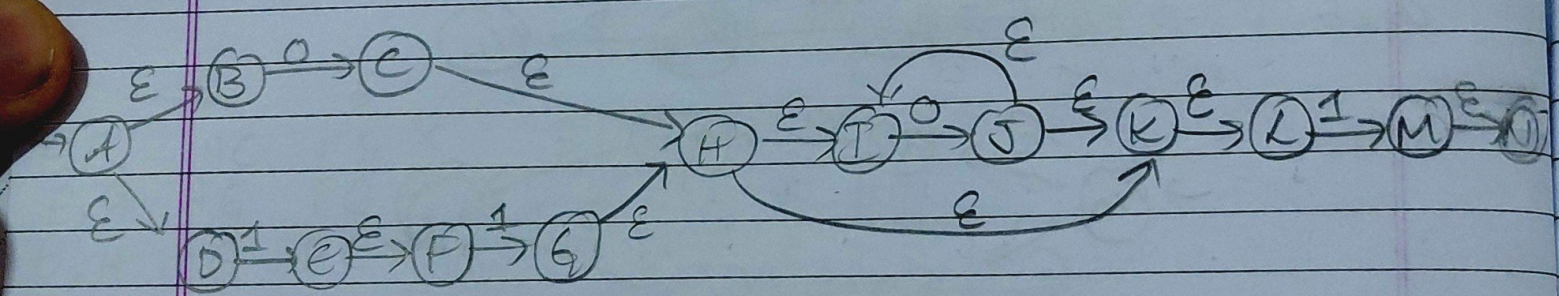
(2) (i) $(a+b)^* bb(a+b)^*$



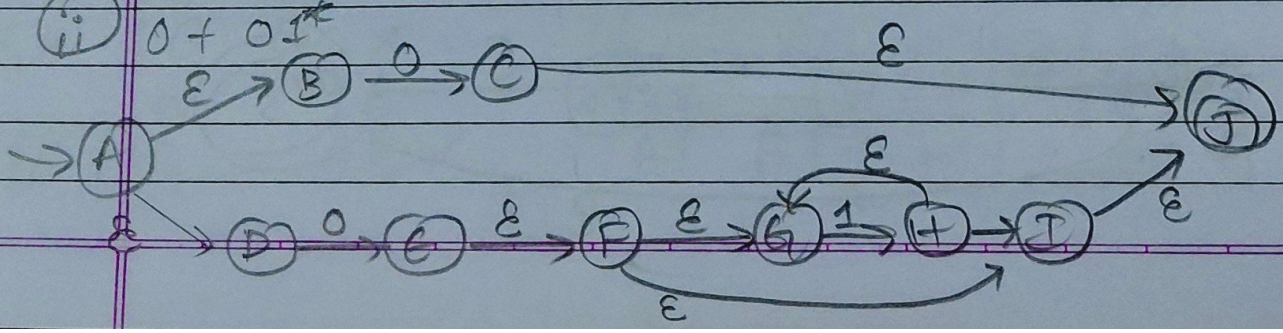
(ii) $ab(a+b)^*$



(3) (i) $(0+11)0^*1$



(ii) $0 + 0 1^*$



4 (a) $ab(a+b)^*$ Same as (ii)

4 (b) Same as (i)

→ $\Sigma\{a, b\}$

7 (i) $a^n b^m$ / $n+m$ is even

$RE = (aa)^* b(bb)^* + a(aa)^*(bb)^*$

(ii) $a^{2n}b^{2m}$ / $n >= 0, m \geq 0$

$(aa)^* (bb)^* \rightarrow a^*a^*b^*b^*$

(iii) $w : n_{ab}(w) \bmod 3 = 0 \quad w \in (a+b)^*$

$RE = ((a+b)^3)^*$ → multiple of 3

$n_a(w) \bmod 3 = 0$

$RE = (b^*ab^*ab^*ab^*)^*$

(iv) a's & b's 5th Symbol from right is a

$RE = (a+b)^* a (a+b)^4$

(v) RE not more than 3 a's

$(\varepsilon+a)(\varepsilon+a)(\varepsilon+a)$ → no more than 3 a's

$RE = b^*(\varepsilon+a)b^*(\varepsilon+a)b^*(\varepsilon+a)b^*$

8 (i) odd length a's & b's

$RE = ((a+b)^2)^* (a+b)$

(ii) 3rd from right a 4th from right is b

$RE = (a+b)^* ba(a+b)^2$

(iii) $0^n 1^m$ / $m \geq 1 \quad n \geq 1 \quad n >= 3$

$m=1 \quad n \geq 3 \quad 000 0^*1$

$m=2 \quad n \geq 2 \quad 00\ 0^*1\ 1$

$m=3 \quad n \geq 1 \quad 0\ 0^*1\ 1\ 1$

$n=1$	$m \geq 3$	$0111 1^*$
$n=2$	$m \geq 2$	$011 1^*$
$n=3$	$m \geq 1$	$01 1^*$

↓

add all to get RE

(9) $L = \{a^n b^m c^p \mid n <= 4, m >= 2, p <= 2\}$

$n=0$ $m \geq 2$ $p=0$ or $p=1$ or $p=2$

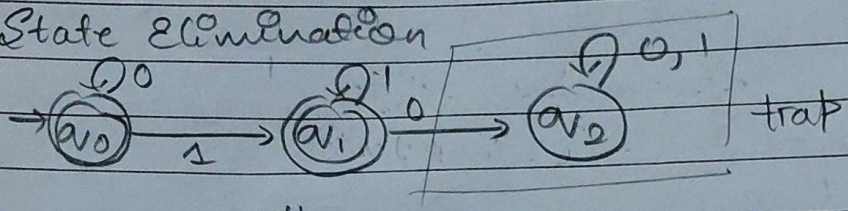
$n=1$ $m \geq 2$

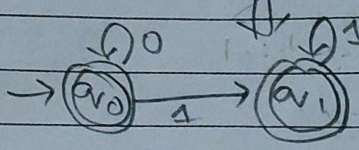
$n=2$ $m \geq 2$

$n=3$ $m \geq 0$

$n=4$ $m \geq 0$

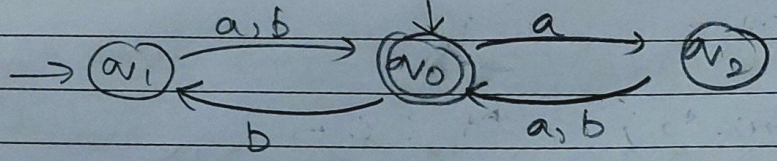
(10) State Elimination





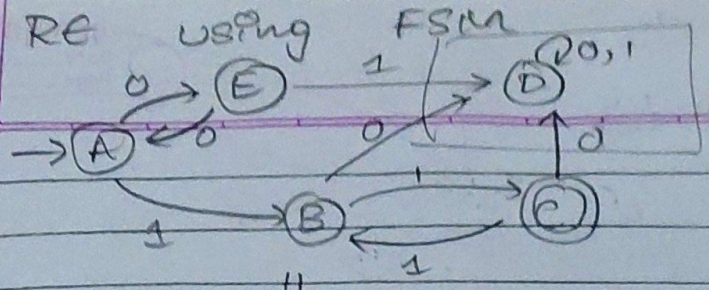
$RE = 0^* + 0^* 1\, 1^* = 0^* + 0^* 1^+$

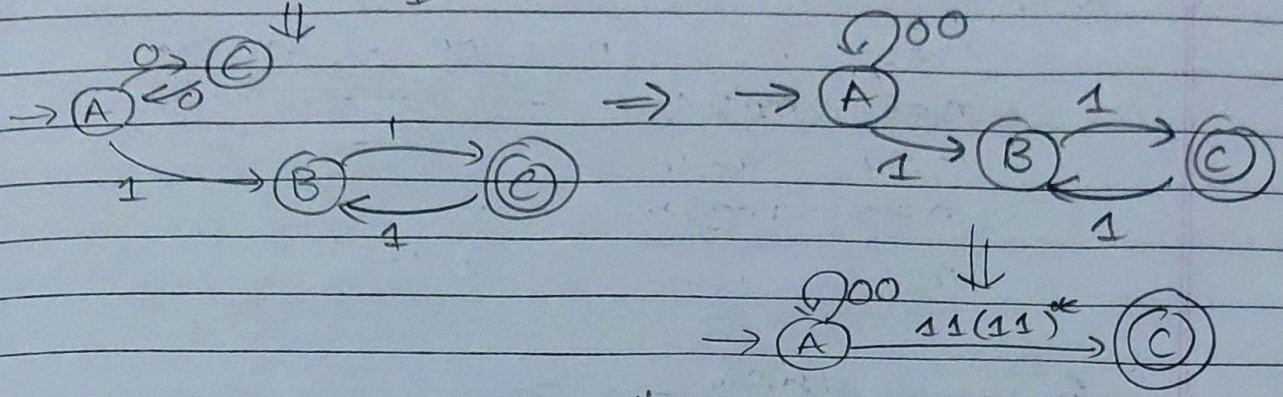
(13)



Can't start with two states NFA.

(14) RE using FSM

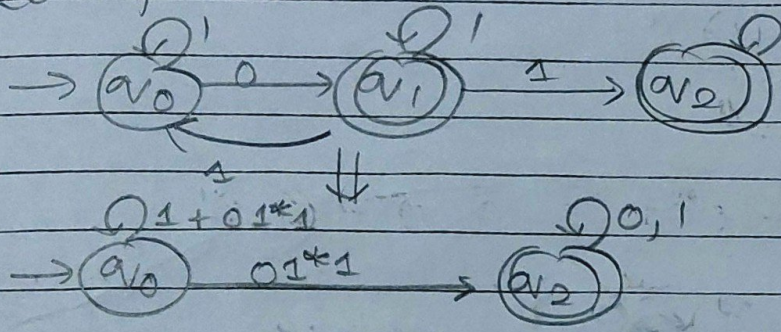




$RE = (00)^* 11(11)^*$

$= (00)^* (11)^+$

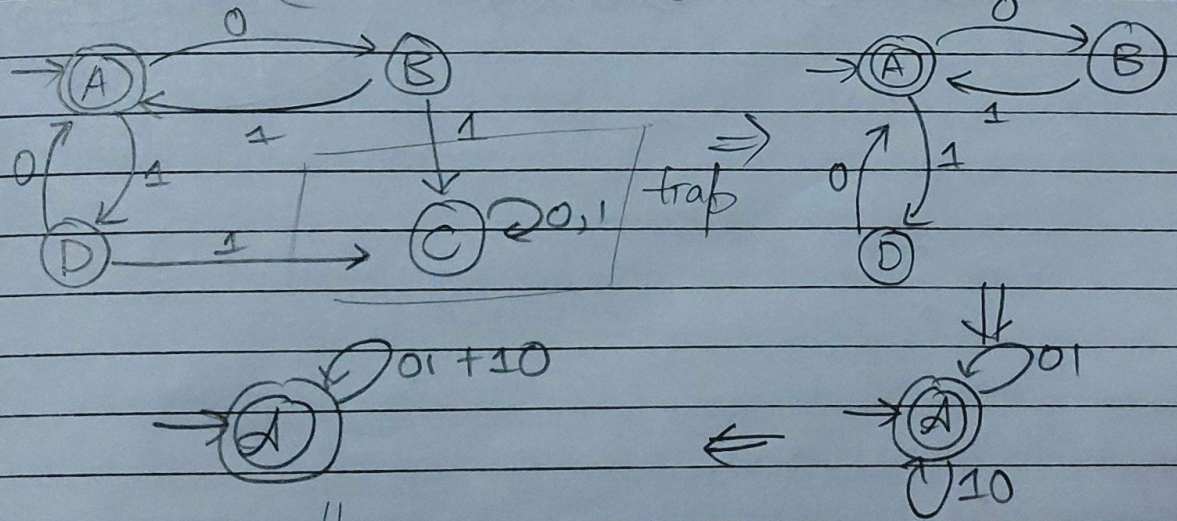
(15) RE for FSM



$RE = (1+01^*1)^* 01^*1 (0+1)^*$

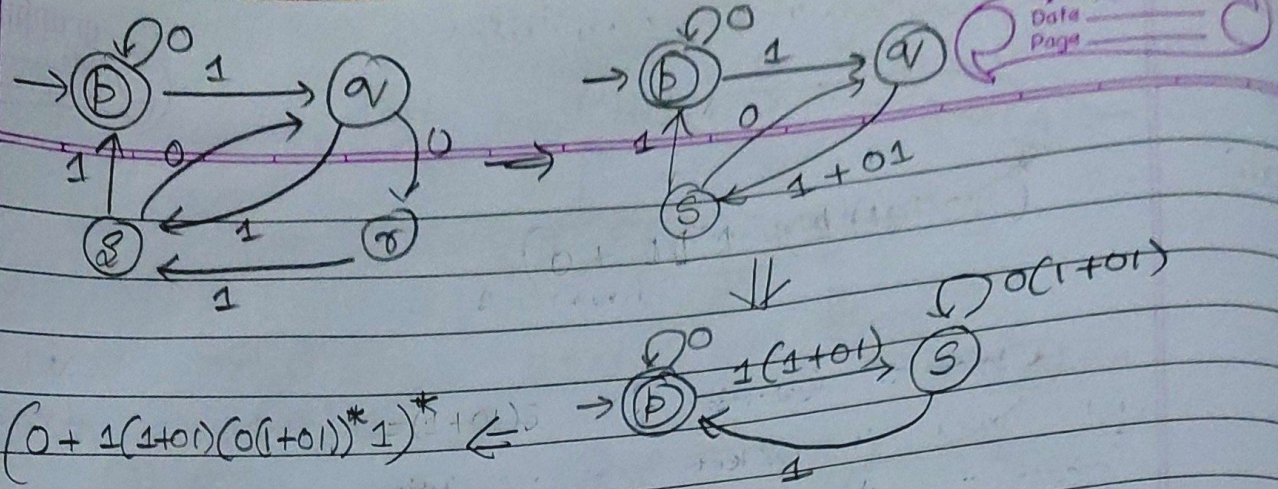
$= (1+01^+)^* 01^+ (0+1)^*$

(16)



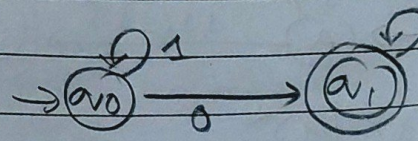
$RE = (01+10)^*$

(17)



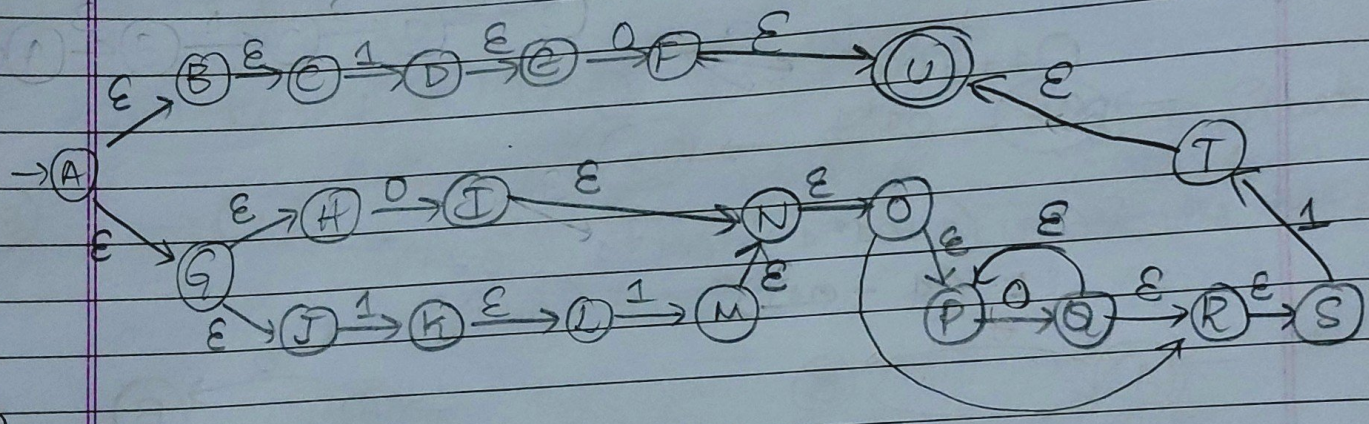
$(0 + 1(1+01)(0(1+01))^*1)^*$

(18)



$RE = 1^*0(0+1)^*$

(19) $RE = 10 + (0+11)0^*1$



(21)

(23) NFSM $RE = (a+b)^*abb$

draw ε NFA

(24) RE for following

(i) $w \in (a,b)^*$ almost 1 a

$RE = b^*(a+\epsilon)b^*$

(ii) dosent end with ba

$RE = \epsilon + a + b + (a+b)^*(aa + ab + bb)$

(iii) 001 ← substring

$RE = (0+1)^*001(0+1)^*$

(iv) w is even $w \in (0,1)$

$RE = ((0+1)^2)^*$

(25) Proof for Every RE is equivalent to FSM

Let R be a regular Expression then ∃ finite automata such that $M = (Q, \Sigma, \delta, q_0, F)$ which accepts Language $L(R)$

by definition ϕ is RE, ε is RE & $\Sigma = a$ is RE

The corresponding machines are

ϕ : → q_0 q_f (no path)

ε : → q_0 —ε→ q_f

$\Sigma = a$: → q_0 —a→ q_f

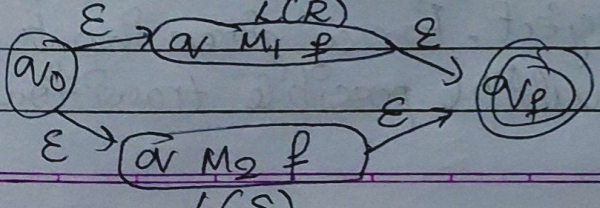
If R & S are RE's then R+S, R.S, R^* are RE's

Schematic Representation of RE of R to Language L(R) is

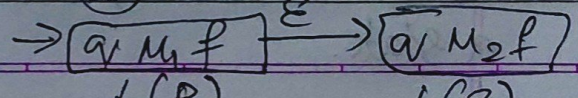
M

→ (q M f) $M = (Q, \Sigma, \delta, q_0, F)$

Case 1 : R = R+S



Case 2 : R = R.S



Case 3: $R = R^*$